

Construction Calculators — Validation Dossier

Formula sources, standards and test proof for every calculator in the workspace

Vendor Workspace (Vendor_Frontend) · Generated 2026-10-01 · Every formula is backed by a cited standard/source and pinned by an automated test (see Vendor_Frontend/tests/calculators.test.jsx).

29

calculators documented

7

Indian Standards cited

20

formula sources with links

8

Cited — constants verified

16

Cited

2

Cited (conservative)

1

Cited — constants verified (conservative)

1

Cited (estimate)

1

Verified

1 · How to verify any number in this document

The test is the proof. Each calculator has a deep-flow test in `tests/calculators.test.jsx` that drives the real UI (open → enter values → press Calculate) and asserts the displayed result against a hand-computed value stated in the test comment. To verify: open the test, read the formula, recompute with a calculator, compare. The full suite (195 tests) runs after every change — if any result drifts from its pinned value, the suite fails.

2 · Indian Standards referenced

STANDARD	WHAT IT GOVERNS IN THE CALCULATORS	LINK
IS 456:2000 (Table 9)	Nominal concrete mixes — M20 = 1:1.5:3, M15 = 1:2:4, M10 = 1:3:6, M7.5 (PCC) = 1:4:8, M25 = 1:1:2	open
IS 2212:1991	Brickwork — 10 mm nominal mortar joint; 500 bricks per m ³	open
IS 1077	Modular brick size 190×90×90 mm	open
IS 2185 Part 3:1984	AAC block nominal sizes — 600/500/400 × 200/250/300 × 100–250 mm	open
IS 2095 Part 1	Gypsum plaster boards — dimensions; market standard board 2400×1200 mm (2.88 m ²)	open
IS 1661:1972	Cement plaster application — mix table and thicknesses (10–15 mm)	open
IS 694	House wire — 90 m coils are the standard market pack	open

3 · About the standards — what they are & how to verify them

IS codes ("Indian Standards") are published by the Bureau of Indian Standards (BIS) — the National Standards Body of India, a statutory body established under the BIS Act 2016 (successor to the Indian Standards Institution, 1947). Each standard is drafted by a technical committee of industry and academic experts (for example CED 2 for concrete, CED 53 for cement-matrix products), opened for public consultation, then published and maintained through a formal lifecycle: revisions (e.g. IS 456 Fourth Revision), amendments between revisions, and periodic reaffirmation.

Indian Standards are voluntary by default, but compliance to many is made mandatory by the Central Government through Quality Control Orders (e.g. cement, TMT steel, electrical wires must carry the BIS/ISI mark). Codes of practice such as IS 456 are mandatory

in effect: the National Building Code of India and municipal building bye-laws reference them, and structural designs in India are submitted and approved against them.

There are different kinds of standards, and our calculators use each kind correctly: Product specifications say what a product must BE (IS 1077 bricks, IS 2185 Part 3 AAC blocks, IS 2095 gypsum boards, IS 694 wires). Codes of practice say HOW to design and build (IS 456 concrete, IS 2212 brickwork, IS 1661 plaster application).

Verify any standard yourself — five steps

1. Find the standard on the BIS "Know Your Standards" portal (bis.gov.in → Know Your Standard, also in the BIS Care app). Searching an IS number shows the IS document, its amendments, gazette notifications, and even the list of licensed manufacturers. (link)
2. Check the official status in the BIS e-Sale store: search the IS number to see Status (Active), Reaffirmed Year, and No. of Amendments. Indigenous standards can be downloaded free of cost after registering. (link)
3. Browse by committee or aspect in the BIS Standards Locator (product specification, code of practice, method of test, dimensions) to confirm what kind of standard you are looking at. (link)
4. Cross-check the exact clause used by a calculator: e.g. IS 456 Table 9 (nominal mixes — M20 = 1:1.5:3), IS 2212 clause 6.4 (10 mm nominal mortar joint), IS 2185 Part 3 clause 3.2 (nominal block sizes). Public copies of many IS codes are mirrored at law.resource.org and archive.org; the BIS store is the authoritative source. (link)
5. Check currency before quoting: confirm the edition is not superseded and note its amendments (IS 456:2000 has 6 amendments; none change Table 9 — the table our calculators use). (link)

CODE	FULL TITLE	COMMITTEE	OFFICIAL STATUS (CHECKED 2026-10-01)	SOURCE
IS 456:2000	Plain and Reinforced Concrete — Code of Practice (Fourth Revision)	CED 2	Active · Reaffirmed 2021 · 6 amendments (latest Amd 6:2024)	official BIS
IS 2212:1991	Code of Practice for Brickwork	CED 30	Active (10 mm nominal joint; bed joints ≤ 12 mm)	official BIS
IS 1077:1991	Common Burnt Clay Building Bricks — Specification	CED 30	Active (modular 190×90×90 mm)	official BIS
IS 2185 (Part 3):1984	Concrete Masonry Units — Part 3: Autoclaved Cellular (Aerated) Concrete Blocks (First Revision)	CED 53	Active · Reaffirmed 2020 · 1 amendment · supersedes IS 5482:1969	official BIS
IS 2095 (Part 1)	Gypsum Plaster Boards — Specification, Part 1: Plain Gypsum Plaster Boards	CED 4	Active (dimensions table for wallboards)	official BIS
IS 1661:1972	Code of Practice for Application of Cement and Cement-Lime Plaster Finishes	CED 4	Active (mix table; 10–15 mm thicknesses)	official BIS
IS 694	PVC Insulated Cables/Wires up to 1100 V — Specification	ETD 1	Active · mandatory BIS certification (ISI mark) for house wires	official BIS

Why these can be trusted

- Statutory body: BIS is established by an Act of Parliament (BIS Act 2016); standards are developed by technical committees with public consultation and a formal revision/amendment/reaffirmation lifecycle.
- Legal force where it matters: many construction products must carry the ISI mark under Quality Control Orders; codes of practice like IS 456 are referenced by the National Building Code and building bye-laws — "as per IS 456" is the standard professional defence.

- Our evidence chain has four links: STANDARD → CLAUSE/TABLE → CONSTANT IN CODE → PINNED TEST. Every calculator row in this dossier names the clause it follows and the test that proves the implementation.
- Honest limits: BIS copies are authoritative but paid (free downloads after registration); free mirrors (law.resource.org, archive.org) may lag the latest amendments — verify currency on the BIS portal; manufacturer datasheets can be imprecise (one AAC supplier cites "IS 2185 Part 3:2015", while BIS lists the 1984 edition as Active/Reaffirmed 2020).
- Estimating-grade: the codes define design and measurement conventions; the calculators apply those conventions for estimation. They are not structural designs and do not replace the project engineer's approved design or the local Schedule of Rates.

4 · Formula sources (well-known calculators & engineering references)

FORMULA / COEFFICIENT	SOURCE
Concrete dry-volume factor 1.54 Voids ~35% when dry ($1 \div 0.65 \approx 1.538 \rightarrow 1.54$)	open source
Plaster dry-volume factor 1.33 Mortar-only factor (no coarse aggregate)	open source
Steel unit weight $d^2/162$ kg/m Full derivation from steel density 7850 kg/m^3	open source
Cement density $1440 \text{ kg/m}^3 \cdot 50 \text{ kg bag} = 0.0347 \text{ m}^3$ Used for all cement-bag conversions	open source
Steel per m^3 of RCC (element thumb rules) Slab 80 · Beam 110–125 · Column 100–160 · Footing 40–80 kg/m^3	open source
Steel per m^2 of built-up area Residential 3.5–4 kg/sqft ($\approx 38 \text{ kg/m}^2$)	open source
Soil swell/bulking factors Common earth ~25% · clay ~30% · sand & gravel ~12% · blasted rock ~50%	open source
Sand / aggregate bulk density Dry sand ≈ 1600 · coarse aggregate $\approx 1500 \text{ kg/m}^3$ (ranges 1450–1750)	open source
Paint coverage (emulsion) Asian Paints PIS: 140–170 sqft/L ($\approx 13\text{--}16 \text{ m}^2/\text{L}$); tool default 10 m^2/L is conservative	open source
Waterproofing spreading rate Dr. Fixit TDS: 3.7–4.2 m^2/L per coat; tool default 1.5 m^2/L is conservative	open source
Formwork plywood sheet + waste $8 \times 4 \text{ ft} = 2.98 \text{ m}^2$ per sheet; +10% waste; sheets = $\text{area} \div 2.98 \times 1.10$	open source
Studs = $\text{ceil}(\text{length} \div \text{spacing}) + 1$ Standard framing count formula (walls & partitions)	open source
Gypsum board $2.88 \text{ m}^2 \cdot \text{studs @400 mm} \cdot \sim 36 \text{ screws/board}$ British Gypsum 300 mm field centres; UK 2400×1200 board	open source
Roofing sheet count via effective cover width sheets = $\text{ceil}(\text{width} \div \text{effective cover width})$ after side lap	open source
AAC thin-bed adhesive $2.7\text{--}3.4 \text{ kg/m}^2$ 40 kg bag covers 160–170 sqft at 3 mm joint	open source

FORMULA / COEFFICIENT	SOURCE
Wall putty coverage Birla White: 1 kg $\approx$ 1.86–2.04 m ² ($\approx$ 0.5 kg/m ² /coat); tool default 0.75 is conservative	open source
Vinyl flooring waste 5–10% (straight lay) Tool default 8% is mid-range	open source
Tile count + waste + whole boxes area $\div$ tile area + waste %, round up to boxes	open source
Board foot = T(in) $\times$ W(in) $\times$ L(ft) $\div$ 12 Standard lumber volume unit	open source
Decking: boards + fasteners take-off Rows $\times$ board runs with 5–10% waste (Omni decking calculator)	open source

5 · Calculator-by-calculator validation

Concrete & Cement 6

Concrete Calculator Cited — constants verified	
Formula	Volume by shape (slab L×W×D, column, cylinder, hollow cylinder) → materials via grade ratio with dry-volume factor. Verified constants: dry factor 1.54 · M10 1:3:6 · M15 1:2:4 · M20 1:1.5:3 · M25 1:1:2 (IS 456 Table 9 — exact match) · w/c 0.6/0.55/0.5/0.45 (at or below IS-456 maxima) · 50 kg bag = 0.035 m ³ .
Sources / standards	IS 456:2000 Table 9 (mixes — exact match verified in code) · Dry-volume factor 1.54 · Cement bag 0.0347–0.035 m ³
Pinned proof	Test pins Volume 125 m ³ for a 5×5×5 m slab; materials flow-verified (cement bags shown).

PCC Calculator Cited	
Formula	Volume = L × W × T; dry volume × 1.54; split by mix (1:4:8, 1:3:6, 1:2:4); cement × 1440 kg/m ³ ÷ 50 kg.
Sources / standards	IS 456 Table 9 (M7.5 = 1:4:8, M10 = 1:3:6) · Dry-volume factor 1.54
Pinned proof	2.5 m ³ @ 1:4:8 → 8.96 bags, 1.244 m ³ sand, 2.488 m ³ aggregate, ₹8,558.

Concrete Column Calculator Cited	
Formula	Volume = L × B × H × count; materials from grade mix table; steel = volume × density (default 120 kg/m ³ , editable).
Sources / standards	IS 456 Table 9 (grades M10–M25) · Steel thumb rule: column 100–160 kg/m ³
Pinned proof	4 columns 230×230 mm × 5 m → 9.95 bags, 133.3 kg steel, ₹15,424.

Concrete Footing Calculator Cited	
Formula	Volume = L × W × T × count; grade mix table; steel default 60 kg/m ³ (editable).
Sources / standards	IS 456 Table 9 · Steel thumb rule: footing 40–80 kg/m ³
Pinned proof	4 footings 5×5×0.5 m → 470.40 bags, 3,150 kg steel, ₹5,24,160.

Concrete Stairs Calculator

Cited

Formula	Stepped prisms (riser $\times$ tread $\div 2 \times$ width $\times$ steps) + waist slab (incline length = $\sqrt{((n \cdot \text{tread})^2 + (n \cdot \text{riser})^2)}$); grade mix table; steel default 80 kg/m ³ .
Sources / standards	IS 456 Table 9 · Standard stair geometry
Pinned proof	5 steps 150/300 mm, 1 m wide $\rightarrow$ 0.364 m ³ , 1.68 m incline, 3.43 bags, ₹4,313.

RCC Formwork Calculator

Cited

Formula	Ply sheets = area $\times$ 1.10 $\div$ 2.97 m ² (8 $\times$ 4 ft sheet); battens = area $\times$ 2.5 m; props = ceil(area $\div$ 1.5); nails = area $\times$ 0.1 kg.
Sources / standards	Sheet 2.98 m ² + 10% waste (picotoolx) · 8 $\times$ 4 ft standard sheet (pinmy.co)
Pinned proof	5 m ² $\rightarrow$ 2 sheets, 12.5 m battens, 4 props, ₹2,675.

Masonry 3

Bricks Calculator

Cited — constants verified

Formula	Wall area $\div$ brick face area including mortar joint (L + joint) $\times$ (H + joint); + wastage %, round up. Verified: mortar joint default 10 mm — exactly the IS 2212 nominal joint.
Sources / standards	IS 2212 (10 mm nominal joint — matches code default) · IS 1077 (190 $\times$ 90 $\times$ 90 modular brick) · infralens brickwork method
Pinned proof	5 $\times$ 3 m wall, Indian Modular 190 $\times$ 90, 10 mm joint, 10% wastage $\rightarrow$ 826 bricks (float-ceil quirk documented in test).

Concrete Blocks Calculator

Cited — constants verified

Formula	Wall area $\div$ block face area (default 406.4 $\times$ 203.2 mm = 16" $\times$ 8" standard CMU) with mortar joint. Verified: mortar joint default 10 mm (IS 2212 nominal); + wastage.
Sources / standards	IS 2185 Part 3 (block nominal sizes) · IS 2572 (concrete masonry units) · 10 mm joint per IS 2212
Pinned proof	Flow verified: default block, 5 $\times$ 5 m wall $\rightarrow$ "Blocks Required" result renders.

AAC Blocks Calculator

Cited

Formula	Blocks = ceil(area $\times$ (1+waste) $\div$ face area); thin-bed adhesive = area $\times$ 3 kg/m ² $\times$ (1+waste).
Sources / standards	IS 2185 Part 3 (600 $\times$ 200 nominal) · Adhesive TDS: 40 kg / 160–170 sqft @ 3 mm = 2.7–3.4 kg/m ²
Pinned proof	5 m ² $\rightarrow$ 44 blocks, 15.8 kg adhesive, ₹2,609.

Steel & Rebar 2

Steel Estimation Calculator

Cited — constants verified

Formula	Quick-area: built-up area × coefficient + wastage × seismic factor. Verified coefficients: residential 38 · commercial 48 · industrial 65 · heavy 60 kg/m ² (cited thumb rules: 3.5–6 kg/sqft ≈ 38–65 kg/m ²). Element-wise: slab 80 · beam 120 · column 100 · footing 60 kg/m ³ (cited ranges: slab 80–100, beam 110–125, column 100–160, footing 40–80). Plus BBS generator mode.
Sources / standards	Thumb rules: 3.5–6 kg/sqft by building type · Element ranges: slab 80–100, beam 110–125, column 100–160, footing 40–80
Pinned proof	Quick-area 125 m ² residential × 38 × 1.05 = 4,987.5 kg — exact-value test.

Rebar / BBS Calculator

Cited

Formula	Unit weight = $d^2 \div 162$ kg/m; per-row weight = unit × cutting length × bars; multi-row totals + wastage.
Sources / standards	$d^2/162$ derivation (from 7850 kg/m ³) · Bar bending schedule practice
Pinned proof	12 mm × 5 m × 5 bars → 0.889 kg/m, 25.0 m, 23.3 kg, ₹1,517.

Earthwork & Aggregates 2

Soil Excavation Calculator

Cited — constants verified

Formula	Volume = L × W × D (or circular); loose volume = volume × swell factor. Verified: swell loose 10 / medium 20 / hard 30 / rock 50 % (cited ranges: sand & gravel 12 · common earth 25 · clay 30 · blasted rock 50). Densities verified: 1600 / 1800 / 2000 / 2400 kg/m ³ (typical soil/rock bulk densities). Vehicle loads by capacity.
Sources / standards	Swell factor tables (Alaska DOT data) · Swell formula: loose = bank × (1 + swell% ÷ 100)
Pinned proof	Flow verified: rectangular 5×5×5 → "Excavation Volume 125 m ³ " + earthwork cost.

Sand & Aggregate Calculator

Cited

Formula	Weight = volume × density (sand 1600 · aggregate 1500 kg/m ³); deliveries = ceil(volume ÷ truck capacity); cost = volume × rate.
Sources / standards	Dry sand ≈ 1602 kg/m ³ (civilead) · Coarse aggregate avg ≈ 1500 kg/m ³ (civilsir)
Pinned proof	5 m ³ sand → 8.4 tonnes (8,400 kg), 1 truck, ₹6,300.

Finishes 6

Plaster Calculator

Cited

Formula	Wet = area × thickness; dry = wet × 1.33; cement = dry × 1 ÷ (1 + sand) × 1440 kg/m ³ ; sand proportional; water = 0.5 × cement.
Sources / standards	IS 1661 (mixes & thicknesses) · Dry-volume factor 1.33 (mortar) · Cement 1440 kg/m ³
Pinned proof	5 m ² @ 12 mm, 1:6 → 0.34 bags (17.2 kg), 0.072 m ³ sand, 8.6 L water, ₹224.

Putty & Primer Calculator

Cited (conservative)

Formula	Putty = area × coats × 0.75 kg/m ² /coat; primer = area ÷ 10 m ² /L; wastage on both.
Sources / standards	Birla White: 1 kg ≈ 1.86–2.04 m ² (≈0.5 kg/m ² /coat) — tool default 0.75 is a conservative upper bound
Pinned proof	5 m ² × 2 coats → 7.88 kg (0.39 bags), 0.53 L primer, ₹449.

Painting Estimator

Cited — constants verified (conservative)

Formula	Paint = paintable area ÷ coverage + wastage; primer with its own coverage; coats configurable. Verified constants: paint 10 m ² /L · primer 11 m ² /L — conservative vs Asian Paints datasheet (140–170 sqft/L ≈ 13–16 m ² /L).
Sources / standards	Asian Paints PIS: 140–170 sqft/L (≈13–16 m ² /L) — tool default 10 is conservative
Pinned proof	Flow verified: wall + ceiling input → “Paint Required” result renders.

Tiles Calculator

Cited

Formula	Tiles = ceil(area ÷ tile area); + waste % (default 10); boxes = ceil(tiles ÷ per-box).
Sources / standards	calculate.co.nz tile method (area ÷ tile + waste, round to boxes)
Pinned proof	5×5 m, 600×600 mm, 10% waste → 77 tiles, 20 boxes, ₹9,240.

Flooring Calculator

Cited — constants verified

Formula	Floor area → tiles by size, boxes, cement & sand for bedding. Verified constants: bedding mortar 20 mm · dry-volume factor 1.33 (mortar) · mix 1:4 (standard tile bedding) · 0.035 m ³ per 50 kg bag · 10 tiles/box default.
Sources / standards	Tile coverage method (calculate.co.nz) · Dry-volume factor 1.33 (mortar) · 1:4 bedding mix (standard practice)
Pinned proof	Flow verified: 5×5 m → “Total Area”, “Tiles Required”, “Cement Bags” render.

Vinyl Flooring Calculator

Cited — constants verified

Formula	Area + 8% wastage → planks/rolls/tiles, adhesive buckets, underlayment. Verified: wastage default 8% (cited range 5–10% straight lay) · adhesive bucket covers 18 m ² (tiles) · underlayment roll 25 m ² .
Sources / standards	Vinyl waste 5–10% straight lay (floorchemdepot) — tool default 8% mid-range
Pinned proof	Flow verified: 5×5 m → “Total Area to Cover 27 m ² (with 8% wastage)” + rolls.

Waterproofing & Envelope 3

Waterproofing Calculator

Cited (conservative)

Formula	Litres = area × coats ÷ coverage (default 1.5 m ² /L/coat); buckets = ceil(litres ÷ 20 L).
Sources / standards	Dr. Fixit Raincoat TDS: 3.7–4.2 m ² /L per coat — tool default 1.5 is conservative
Pinned proof	5 m ² × 2 coats → 7.0 L, 1 bucket, ₹1,750.

Roofing Calculator

Cited

Formula	Slope area = $\text{plan} \div \cos(\text{atan}(\text{slope}\%))$; sheets = $\text{ceil}(\text{slope area} \times (1 + \text{waste}) \div \text{sheet area})$; fasteners = $\text{ceil}(\text{area} \times \text{screws}/\text{m}^2 \times (1 + \text{waste}))$.
Sources / standards	Sheet count via effective cover width (knowsteel) · Corrugated sheet layout calculator (codingace)
Pinned proof	5×5 m @ 20% slope, 3×1.05 m sheets → 25.50 m ² , 9 sheets, 215 screws, ₹8,295.

Retaining Wall Calculator

Cited

Formula	Stem volume ($L \times H \times \text{thickness}$) + base volume ($L \times W \times T$); grade mix table; steel default 80 kg/m ³ (editable).
Sources / standards	IS 456 Table 9 · Steel thumb rules (footing/RCC ranges 40–80 kg/m ³)
Pinned proof	5 m × 5 m, 200 mm stem → 68.21 bags, 609 kg steel, ₹85,899.

Partitions & Ceilings 1

Drywall / Partition Calculator

Cited

Formula	Boards = $\text{ceil}(\text{area} \times (1 + \text{waste}) \div 2.88 \text{ m}^2)$; studs = $\text{ceil}(\text{length} \div \text{spacing}) + 1$; screws = $\text{area} \times 12/\text{m}^2$; joint compound = $\text{area} \times 0.5 \text{ kg}/\text{m}^2$.
Sources / standards	Board 2400×1200 = 2.88 m ² · studs @400 mm · ~36 screws/board (British Gypsum) · IS 2095 (board dimensions)
Pinned proof	5×5 m @ 400 mm → 10 boards, 14 studs, 315 screws, 13.1 kg compound, ₹10,398.

Timber & Framing 4

Decking Calculator

Cited

Formula	Rows = $\text{ceil}(\text{width} \div (\text{board width} + \text{gap}))$; boards/row = $\text{ceil}(\text{length} \div \text{board length})$; boards with waste; screws = $\text{area} \times 25/\text{m}^2 \times (1 + \text{waste})$.
Sources / standards	Decking take-off method (Omni decking calculator) · Waste 5–10% (flooring waste guides)
Pinned proof	5×5 m, 140 mm boards + 5 mm gap, 3 m lengths → 74 boards, 657 screws, ₹35,271.

Framing Calculator

Cited

Formula	Studs = $\text{ceil}(\text{length} \div \text{spacing}) + 1$; stud timber = studs × height; plates = 2 × length; stock lengths = $\text{ceil}(\text{linear} \div 3 \text{ m})$.
Sources / standards	Stud count formula: $\text{ceil}(L \div \text{spacing}) + 1$ (codingace) · 400 mm standard stud spacing
Pinned proof	5 m × 5 m @ 400 mm → 14 studs, 73.5 m + 10.5 m timber, 29 stock lengths, ₹5,220.

Lumber Calculator

Cited

Formula	Board feet = $T(\text{in}) \times W(\text{in}) \times L(\text{ft}) \times \text{pieces} \div 12$; CFT = $\text{BF} \div 12$; cost = $\text{CFT} \times \text{rate}$.
Sources / standards	Board foot standard definition (Omni board foot calculator)
Pinned proof	5 pieces 1×6 in × 8 ft → 20.00 BF, 1.67 cft, ₹4,167.

Roof Truss Calculator

Cited (estimate)

Formula	$\text{Trusses} = \text{ceil}(\text{length} \div \text{spacing}) + 1$; $\text{rise} = \text{span} \times \text{pitch\%} \div 2$; $\text{rafter} = \sqrt{(\text{span}/2)^2 + \text{rise}^2}$; $\text{timber per truss} = 2 \times \text{rafter} + \text{span} + 0.4 \times \text{span}$ (web estimate); + wastage.
Sources / standards	Truss take-off & timber tools (BuildingClub Pro timber suite) · Standard gable geometry
Pinned proof	5 m span, 5 m length @ 30% pitch, 1.2 m spacing → 6 trusses, 12.22 m per truss, 77.0 m timber, ₹13,858.

MEP 1

Electrical Wiring Estimator

Cited — constants verified

Formula	$\text{Wire per point} = \text{distance} \times 2 \text{ conductors (live + neutral)} \times 1.10 \text{ slack}$; $\text{coils} = \text{ceil}(\text{total} \div 90 \text{ m})$; $\text{conduit} = \text{run} \times 80\% \times 1.10 \div 3 \text{ m pipes}$; load points by type. Verified: 10% slack, 90 m coil, 80% conduit share, 3 m pipe.
Sources / standards	90 m coil standard pack (KEI/Polycab, IS 694) · Conduit 3 m pipe standard
Pinned proof	Flow verified: 5 points → "Total Points", "Wire Required 145.2 m (2 coils)", conduit.

Cost & Logistics 1

Freight Cost Calculator

Verified

Formula	$\text{Cost} = \text{distance} \times \text{rate} + \text{fuel surcharge\%} + \text{tolls} + \text{handling fee}$.
Sources / standards	Arithmetic (no standard required)
Pinned proof	100 km × ₹10 + 10% fuel + ₹50 toll + ₹25 handling = ₹1,175.00 — exact-value test.

6 · Scope & caveats

- These are **estimating-grade** tools — they follow the cited conventions but are not structural designs or legal measurements (no IS-1200 deduction schedules, no IS-10262 design mixes).
- **Rates are editable market seeds** (e.g. cement ₹400/bag, sand ₹1,200/m³, steel ₹65/kg) — replace with your city SOR / current quotes before tendering.
- Defaults marked "*Cited (conservative)*" deliberately over-order versus the manufacturer datasheet so material is not under-ordered; all are editable in the UI.
- Source links are a snapshot as of 2026-10-01 and should be re-checked if standards are revised.